

WayLens

A Generative AI Based Indoor Classroom Navigation Assistant for the Visually Impaired

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1. Background

Independent indoor navigation remains a significant challenge for visually impaired individuals, particularly within educational institutions where classrooms, laboratories, and administrative offices are distributed across complex building layouts. Unlike outdoor environments, indoor spaces lack reliable positioning technologies such as GPS, making it difficult for a visually impaired person to determine their current location and reach a desired destination without external assistance. This limitation directly affects accessibility, independence, and inclusion in academic environments.

According to the World Health Organization, over 250 million people worldwide experience some form of visual impairment. In educational settings, this population includes students, faculty, and visitors who must independently locate classrooms and facilities on a daily basis, which makes intelligent, low-cost assistive navigation technology a genuine and pressing need.

2. Research Gap

Existing indoor navigation solutions predominantly rely on GPS, Bluetooth beacons, RFID infrastructure, wearable devices, or static tactile maps.

- GPS based navigation: Ineffective indoors and cannot resolve location at the level of an individual classroom.
- Beacon, RFID, or sensor-based infrastructure: Requires costly installation and ongoing maintenance across every building, which limits scalability across an institution.
- Wearable devices such as smart glasses: Introduce usability, comfort, and safety concerns, particularly where hardware is mounted near the eyes.
- Static tactile maps and Braille signage: Provide only fixed information and cannot generate dynamic, destination specific guidance.

Beyond these infrastructure limitations, existing approaches rarely integrate visual scene understanding with natural language reasoning to generate adaptive, human understandable navigation instructions. There remains a clear gap in developing an intelligent, infrastructure light indoor navigation system that can interpret visual information and deliver context aware, conversational guidance.

3. Problem Statement

There is currently no intelligent, multimodal system capable of identifying a visually impaired user's current indoor location from a captured scene and generating conversational, context aware navigation instructions toward a specified classroom. Existing solutions either lack accurate indoor localization or fail to provide guidance that adapts to the user's immediate surroundings. This creates the need for a Generative AI driven framework that combines visual perception, structured building knowledge, and language-based reasoning to enable independent indoor navigation, delivered in a form that does not depend on the visually impaired user having to see, aim, or frame the input themselves.

4. Proposed Methodology

WayLens proposes a multimodal Generative AI framework for indoor classroom navigation. The system accepts an image of a corridor or classroom entrance together with a destination classroom number provided through voice or text. A Vision Language Model integrated with optical character recognition analyzes the image to recognize classroom identifiers, textual signage, and environmental landmarks, thereby estimating the user's current location.

The estimated location is mapped to a predefined digital representation of the building, in which classrooms and landmarks are organized as interconnected spatial entities. A spatial reasoning module determines the optimal navigation path between the current and destination locations. A Large Language Model then synthesizes the visual context and route information to generate conversational navigation instructions, which are converted to speech through a text to speech module so that guidance is delivered entirely by voice.

The research contribution lies in the integration of multimodal perception, knowledge grounded spatial reasoning, and language model-based instruction generation, rather than in the hardware used for image acquisition.

5. Current Scope: Software Prototype

A visually impaired user cannot be expected to independently aim a camera or frame a photograph, since the system is intended to assist them precisely because they cannot see. This is an intended usability constraint, not an implementation detail to resolve later. The present phase of this research therefore implements and validates only the Generative AI pipeline. Image capture is provided through a camera or gallery upload as a trial and demonstration mechanism, allowing the perception, reasoning, and instruction generation components to be tested and evaluated independently of any specific hardware. This method of input is a testing convenience for the current research phase and is not proposed as the intended mode of interaction for an end user.

6. Future Scope: Product Integration

Beyond the current research phase, the framework is designed to extend into a standalone handheld navigation device that removes the need for the user to manually capture an image. The proposed form factor is a handheld stick fitted with a camera and a microcontroller, capable of continuous or on demand scene capture without requiring the user to aim or hold up a phone. A wearable form such as smart glasses was considered and deliberately ruled out at this stage due to the physical risk and liability associated with any device mounted near the eyes. The handheld stick offers a safer, lower risk path to a physical prototype while preserving the same underlying software pipeline. Subject to time availability before submission, an ESP32-CAM based implementation will be explored as an initial hardware trial toward this direction.

7. Objectives

- Develop a multimodal perception pipeline capable of estimating the user's current location from a single indoor image.
- Construct a structured building knowledge representation for classroom level navigation.
- Integrate Vision Language Models with Large Language Models for context aware route reasoning.
- Generate natural language navigation instructions suitable for voice based interaction.
- Demonstrate an infrastructure light indoor navigation framework that can extend toward a handheld product for visually impaired users.

8. Expected Outcomes

- Accurate identification of classroom locations from captured corridor or classroom images.
- Reliable estimation of the user's current position using multimodal scene understanding.
- Knowledge grounded route planning based on building layout information.
- Human like navigation instructions generated through a Large Language Model.
- Voice based indoor navigation that promotes independent mobility within academic environments.
- A validated software framework with a defined path toward a handheld hardware prototype for practical deployment.

9. Novelty

Unlike conventional indoor navigation systems that depend on GPS, Bluetooth beacons, RFID infrastructure, or wearable technologies, WayLens AI introduces a multimodal Generative AI framework that integrates visual scene understanding, structured building knowledge, spatial reasoning, and natural language generation to provide conversational indoor navigation assistance from a single captured image. The approach also addresses a usability constraint that is often overlooked in prior work, namely that a visually impaired user cannot be relied upon to capture their own image, and proposes a handheld, automatic capture device as the intended deployment form rather than treating image acquisition as a solved or trivial input step.